

SCALES IN THE UNIVERSE = from 1m to the size of the Universe.

1 m is about 1 yard and 1km is 1,000 m, 1 light year is 10^{13} km



100m x 100m=San Marco



1000m x 1000m = Venice (1km x 1km)
10km x 10km the whole areal

Our place in the Universe

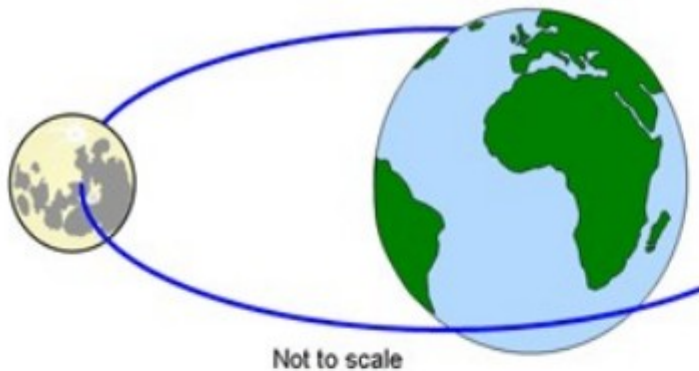
Distance and time scale



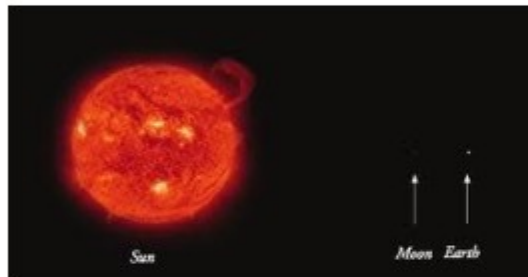
1000kmx 1000km
(distance Chicago-NY)



10,000kmx 10,000km
The Earth (R=6000km)



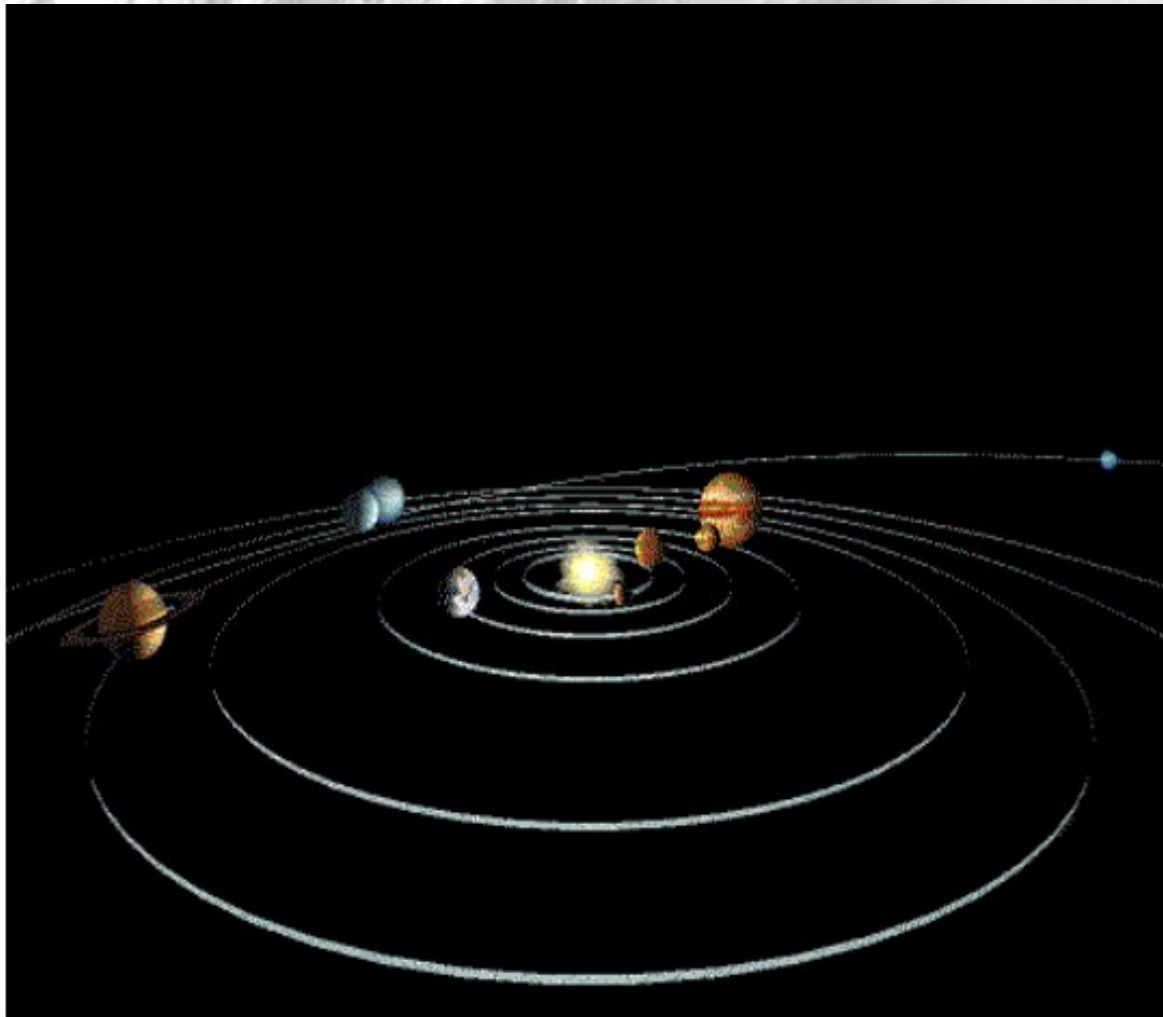
500,000 km x 500,000 km to cover the Moon orbit. (384, 403 km)



**Distance Earth-Sun =
150 million km = 1 AU
(astronomical unit)**

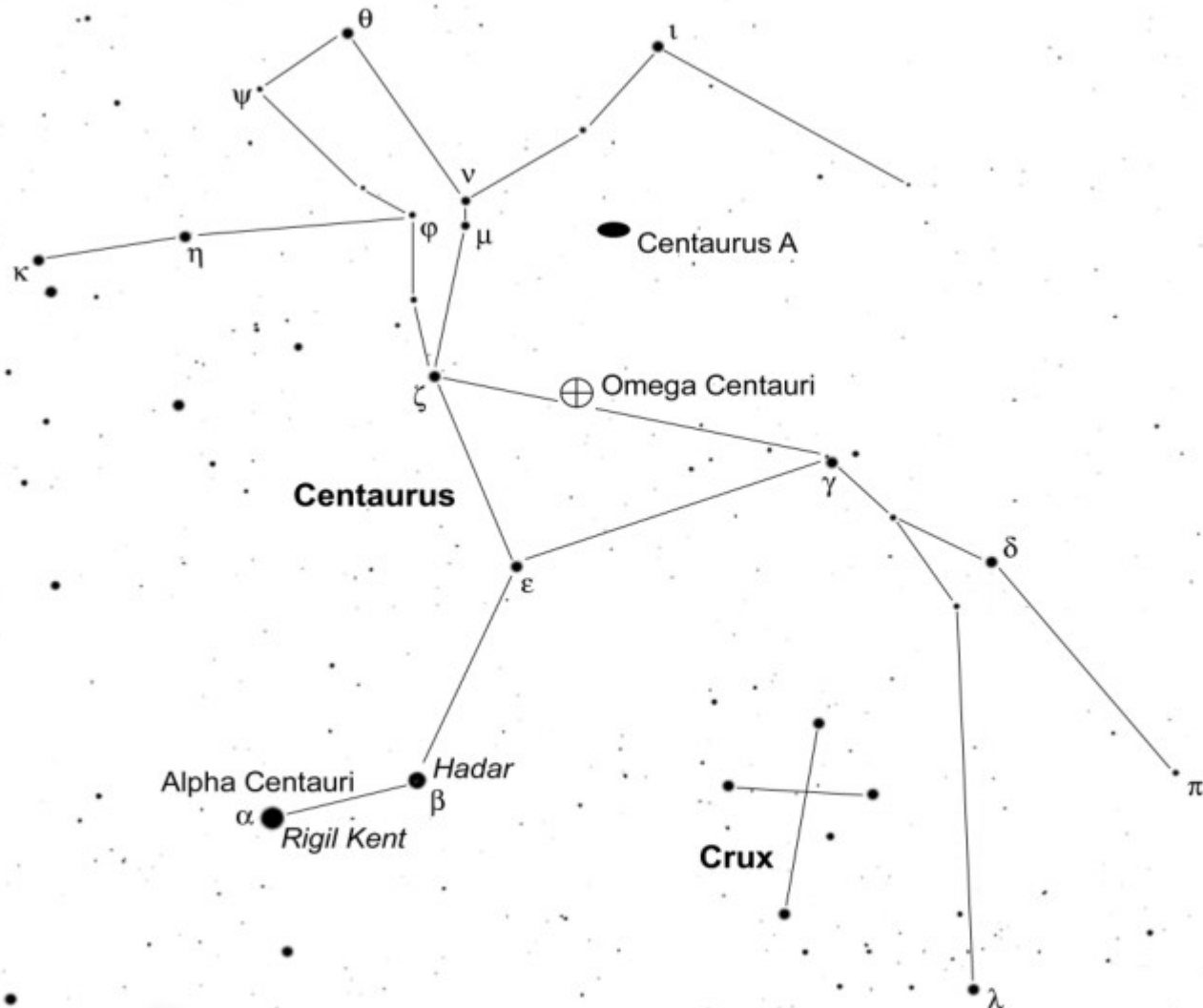
Sun-Planet Distances

Planet	Copernicus	Modern
Mercury	0.38	0.387
Venus	0.72	0.723
Earth	1.00	1.00
Mars	1.52	1.52
Jupiter	5.22	5.20
Saturn	9.18	9.54



1 billion km x 1 billion km to cover the
Solar system (including Pluto !!)



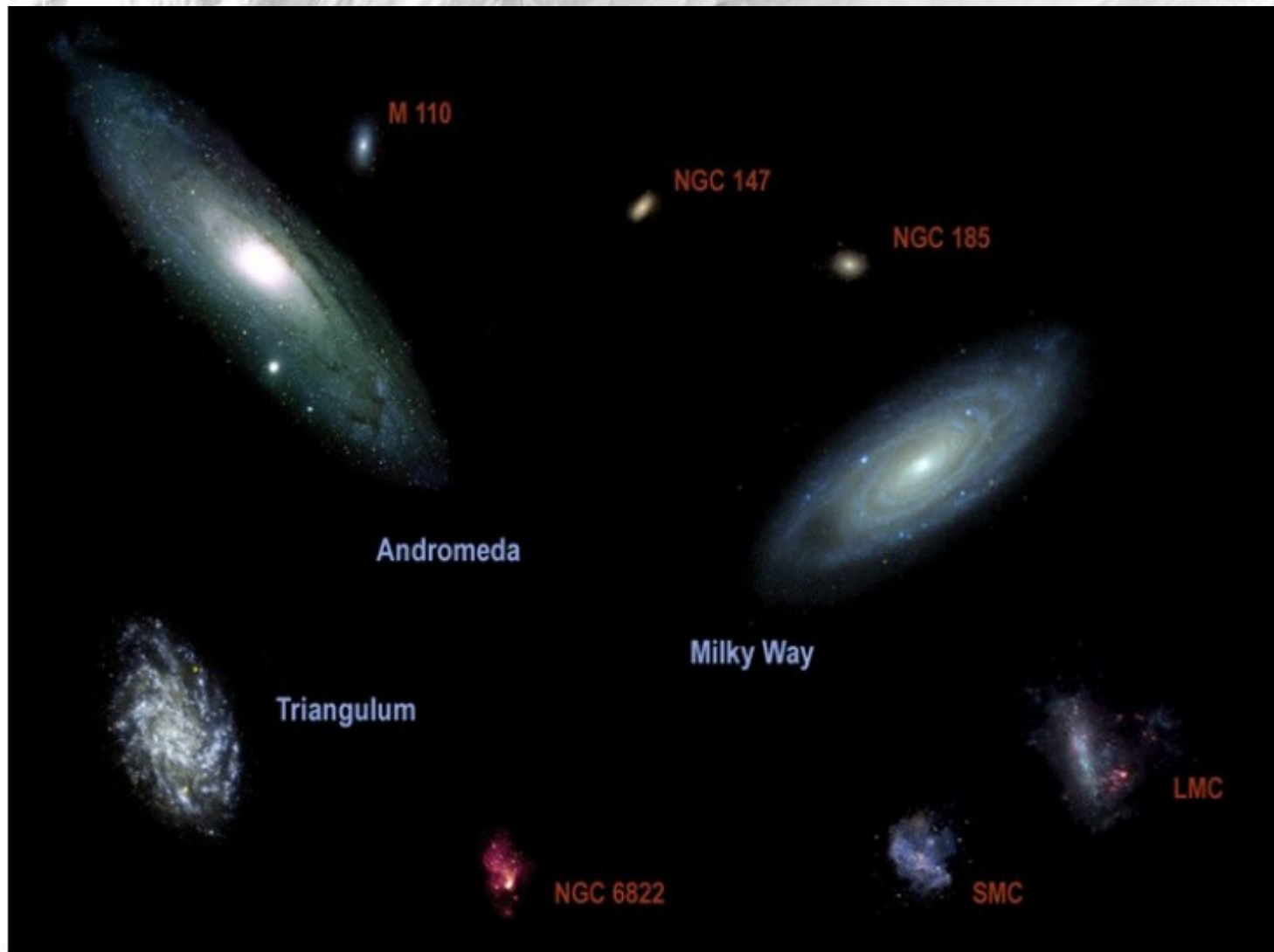


- Closest star is proxima centauri at about **4.3 light years** away. Part of a multiple system Alpha centauri (24AU between 2 binary stars (Uranus) , 15000 AU with proxima)
http://en.wikipedia.org/wiki/Alpha_Centauri

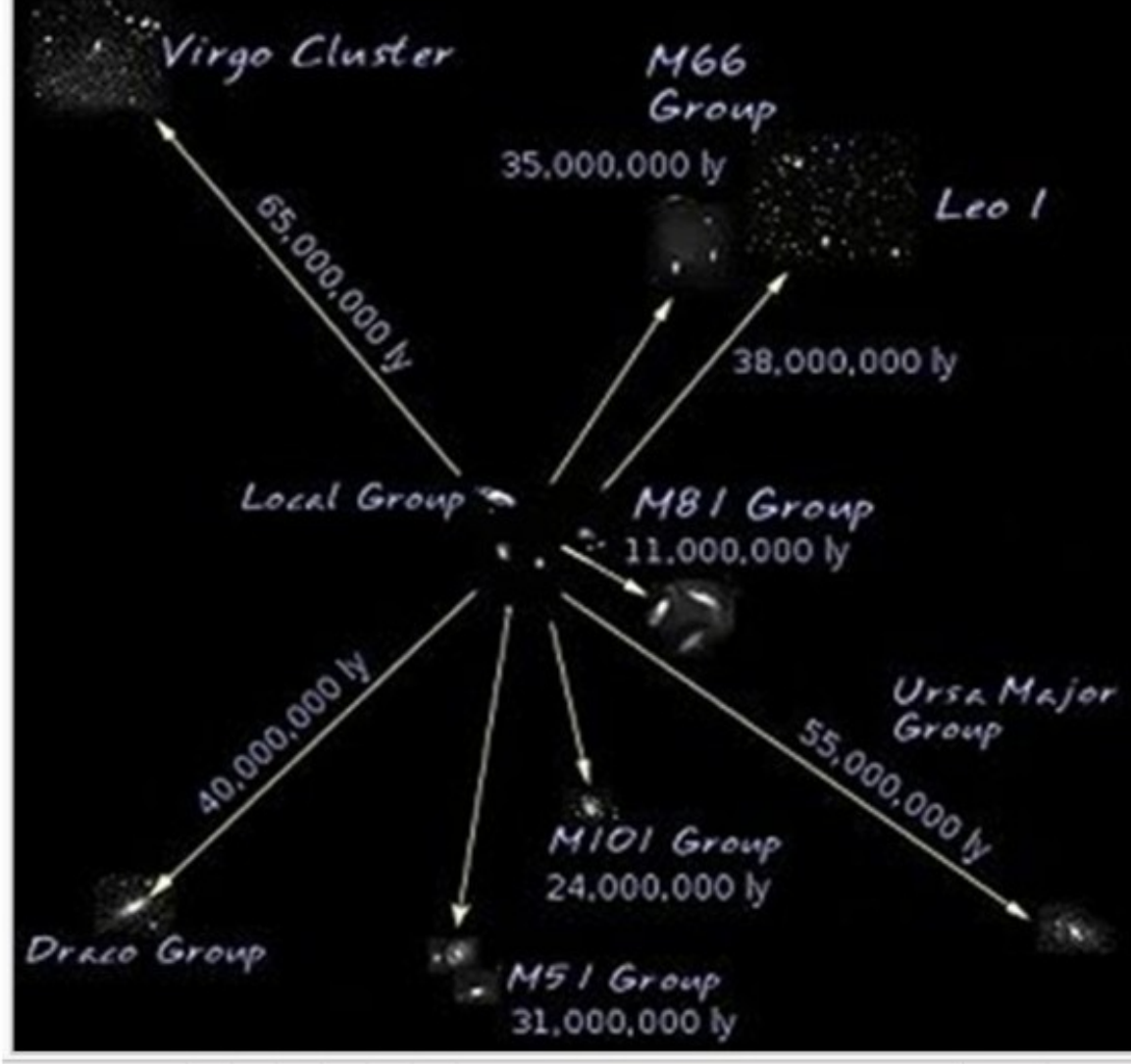


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Milky way = 100,000 light years across.



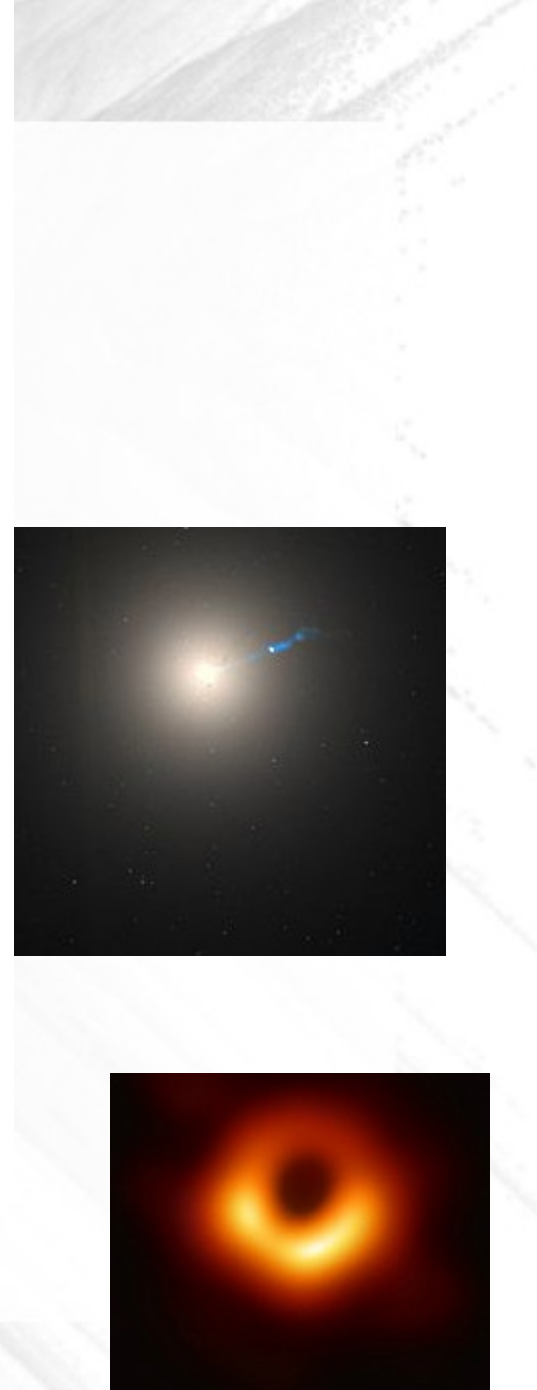
Andromeda Galaxy = 2.5 million light years away.

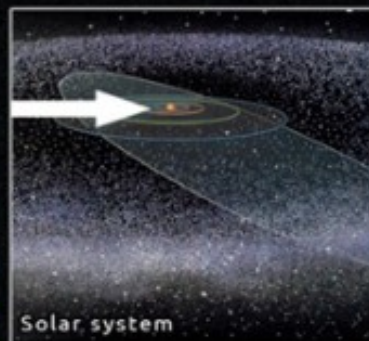


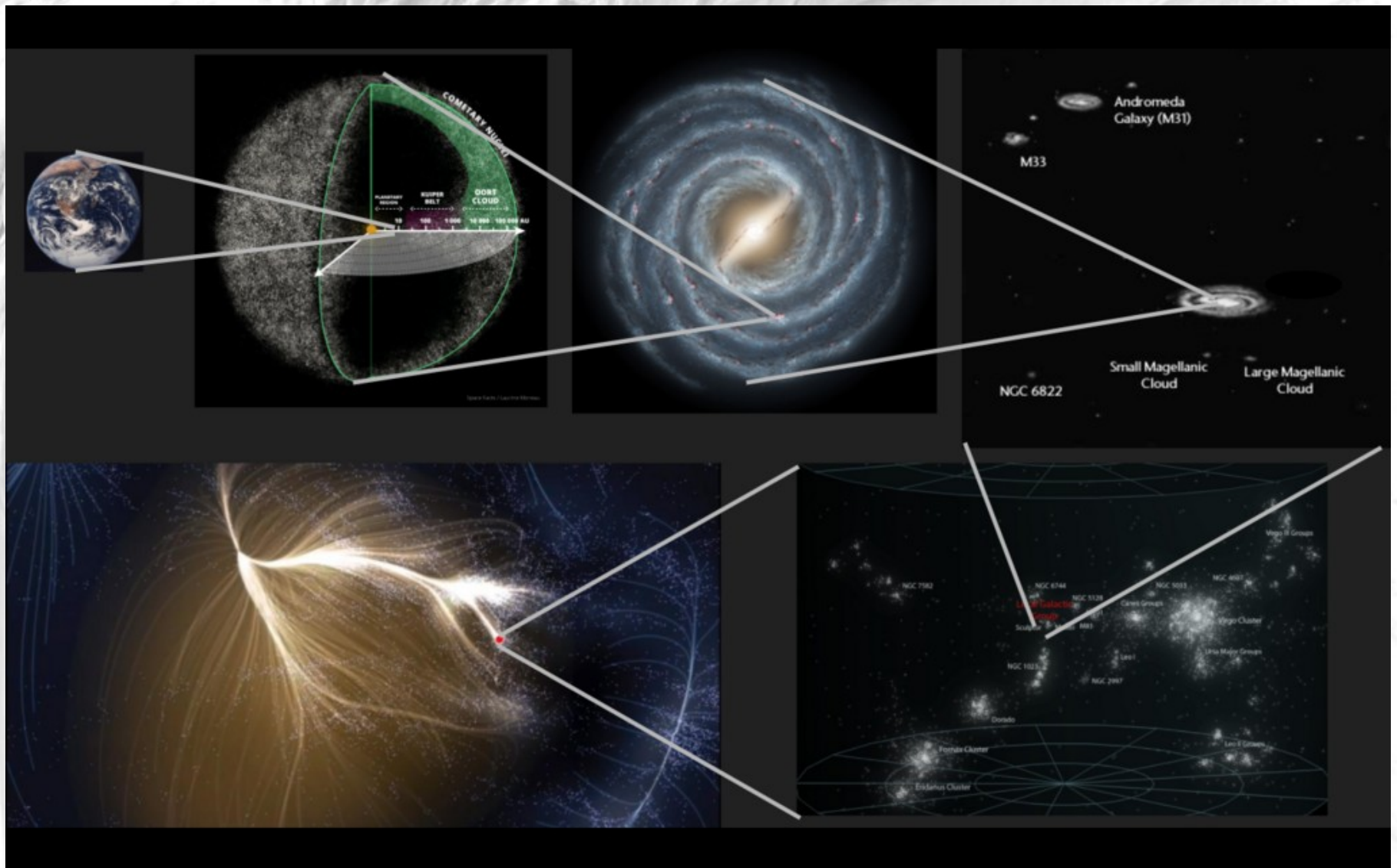
Virgo group = 60 million light years

The monstrous M87 is the dominant member of the neighboring Virgo cluster of galaxies, which contains some 2,000 galaxies. Discovered in 1781

In 2017 → 1st pic of a BH . Even horizon telescope



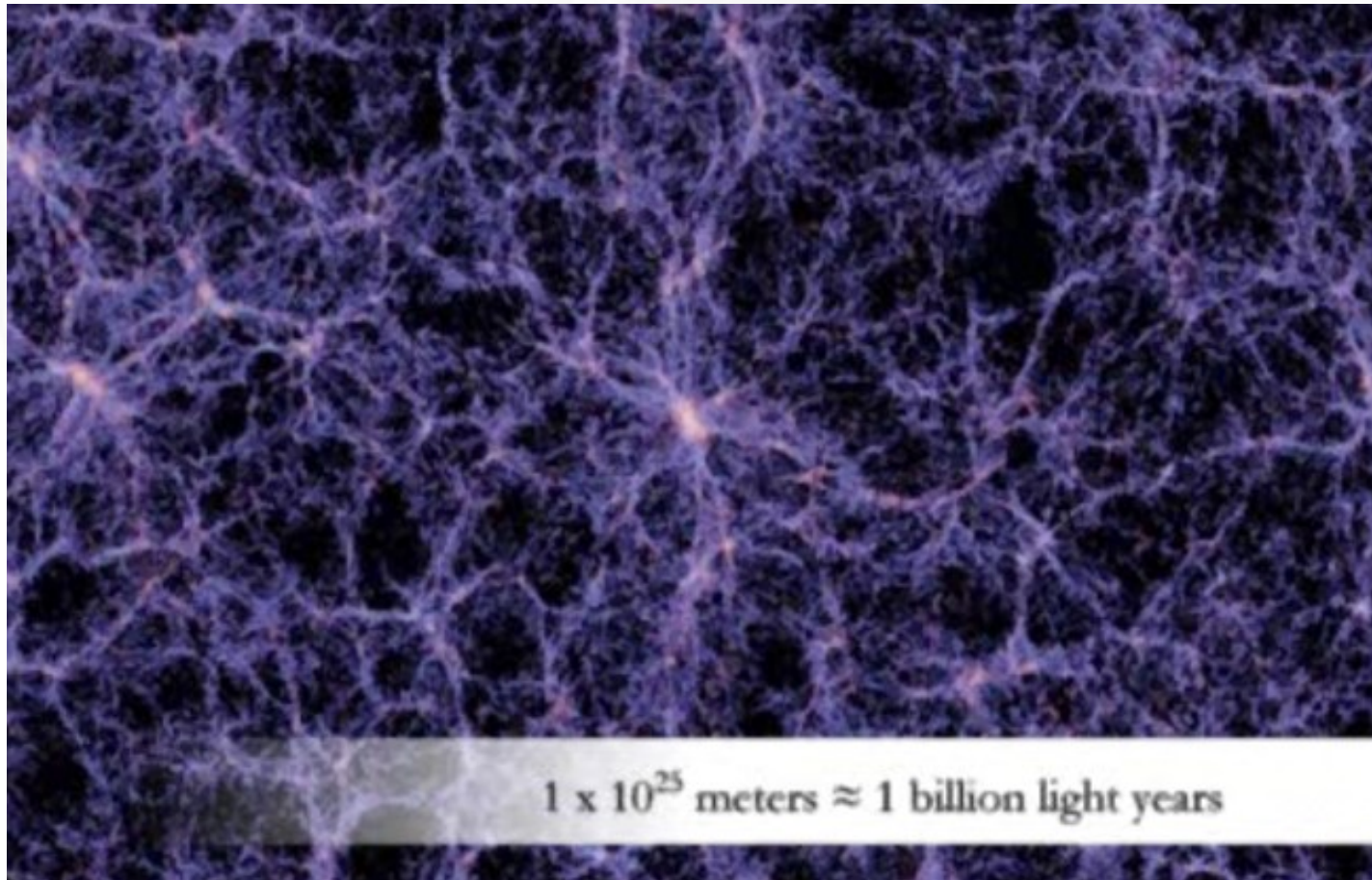




<https://quizlet.com/220003127/astronomy-intro-our-place-in-the-universe-diagram/>

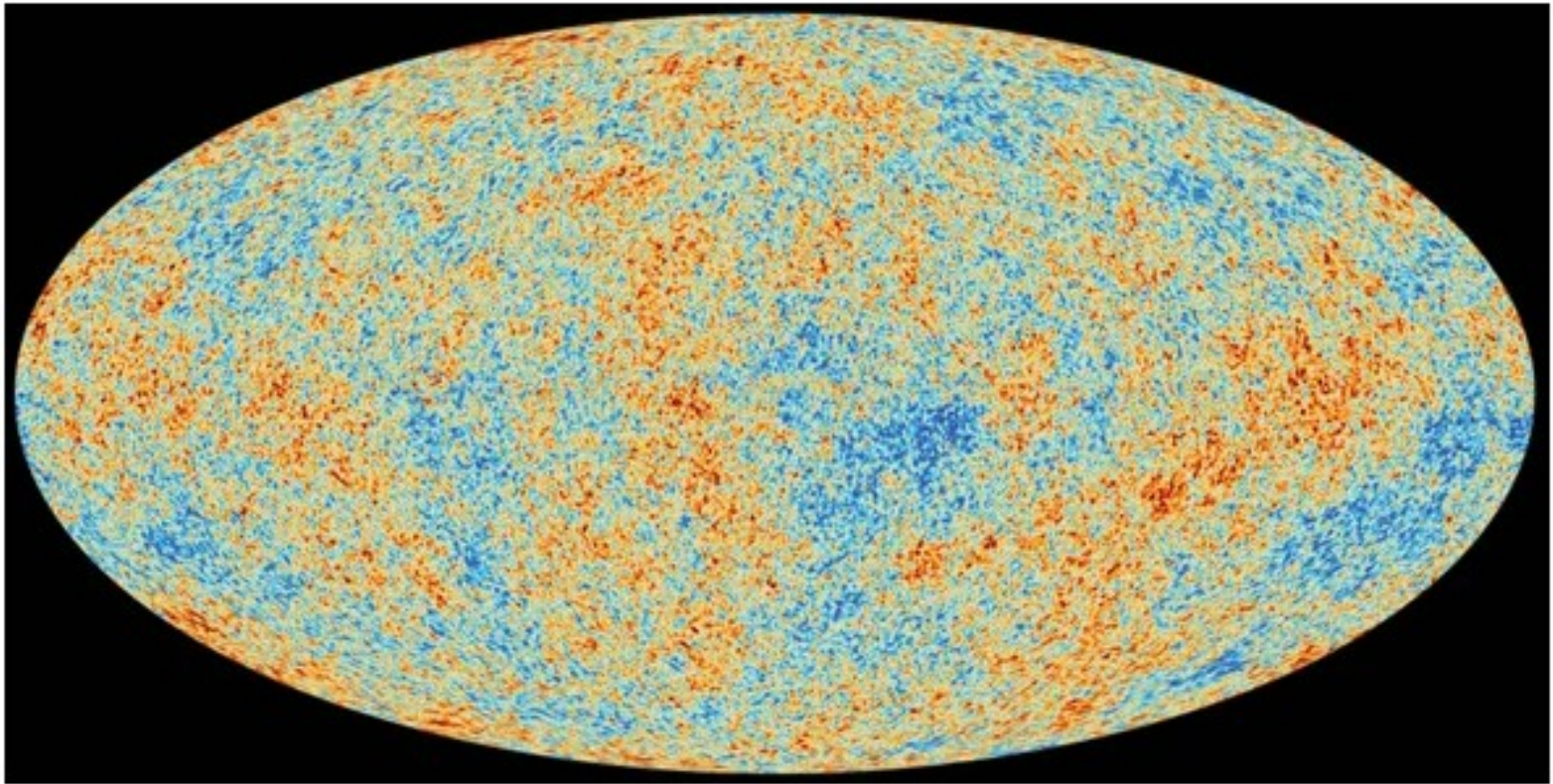


1×10^{24} meters $\approx$ 100 million light years



<https://newatlas.com/dark-matter-filaments-found/23281/>

The universe



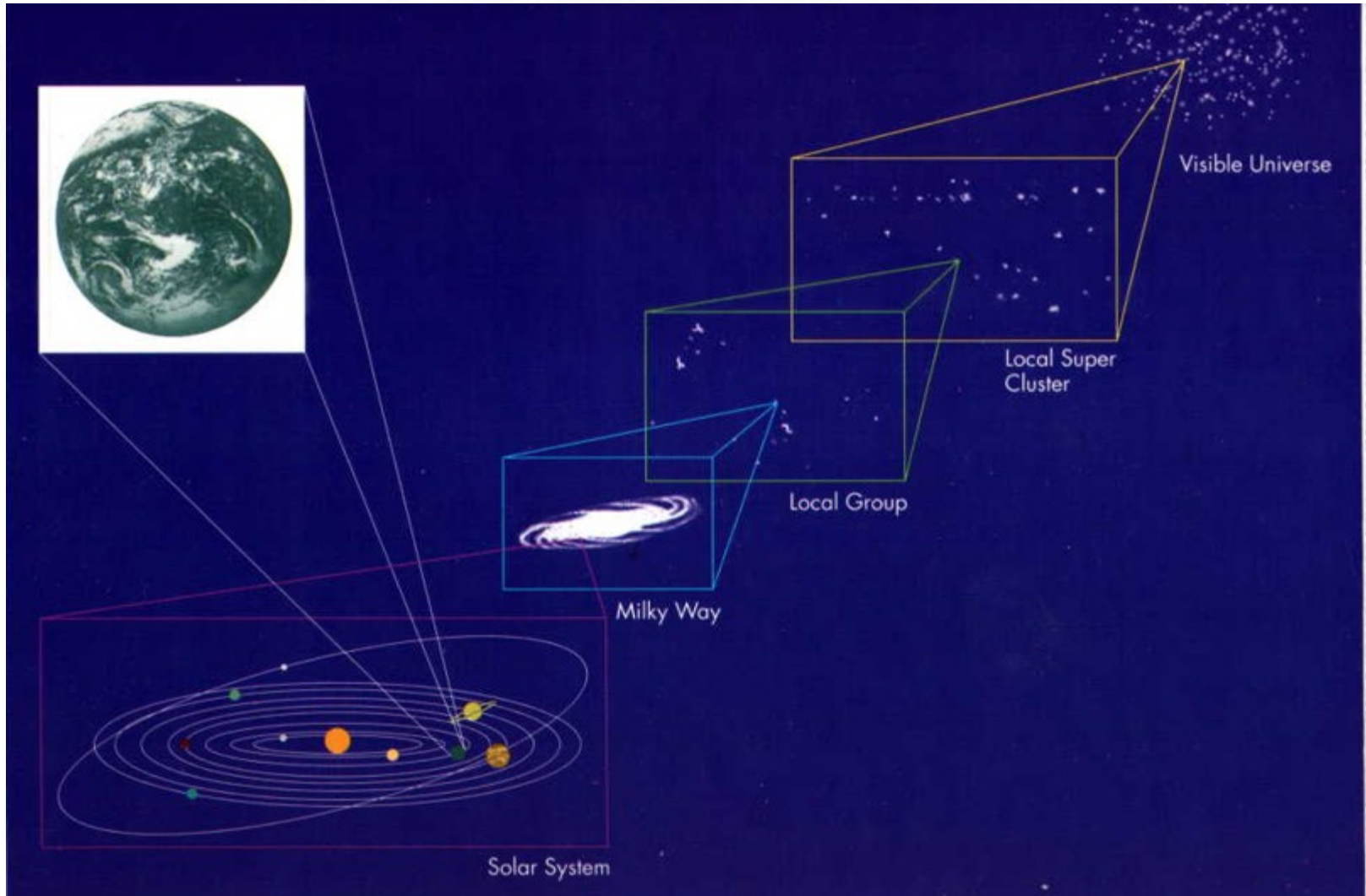
Emitted just 380,000 years after the big bang, the cosmic microwave background is the oldest observable light in the universe. By studying patterns imprinted upon this ancient light, scientists can surmise the universe's age, size and large-scale structure. Credit: ESA and Planck Collaboration

Scale in the universe

→ see our place video

https://en.wikipedia.org/wiki/Alpha_Centauri

→ see known universe



Metric vs. British Units

1 meter (m) = 39.37 inches

1 kilometer (km) = 1000 m $\approx$ 0.62 mile

1 centimeter (cm) = 0.01 m $\approx$ 0.39 inch

453.6 grams (g) = 1 pound

1 kilogram (kg) = 1000 g $\approx$ 2.2 pounds

1 A.U. 150,000,000 km $\rightarrow$ how much in miles?
 $\rightarrow$ Scientific notation?

Remember: 10K run = 6.2 miles about
5K run = 3.1 miles

Light-year

- The **distance** light can travel in 1 year
- About 10 trillion kilometers (6 trillion miles)



1 mile = 1.6km

186,000 miles/s $\rightarrow$? km

300,000 km/s

Show video – speed of light

How long for light from the Moon to reach us?

Light travels at a finite speed (300,000 km/s).



382,500 kilometers

$$(\text{distance}) = (\text{speed}) \times (\text{time})$$

$$d = v \quad t$$

$$\frac{d}{v} = t$$

$$t = \frac{d}{v}$$

$$t = \frac{3.84 \times 10^5 \text{ km}}{3 \times 10^5 \text{ km/s}}$$

How far is a light-year?

1 light-year = (speed of light) $\times$ (1 year)

$$= \left(300,000 \frac{\text{km}}{\cancel{s}} \right) \times \left(\frac{365 \cancel{\text{days}}}{1 \cancel{\text{yr}}} \times \frac{24 \cancel{\text{hr}}}{1 \cancel{\text{day}}} \times \frac{60 \cancel{\text{min}}}{1 \cancel{\text{hr}}} \times \frac{60 \cancel{s}}{1 \cancel{\text{min}}} \right)$$
$$= 9,460,000,000,000 \text{ km}$$

In scientific notation this is about ? In km ?

1 mile = 1.6 km

Convert to miles and trillion of miles

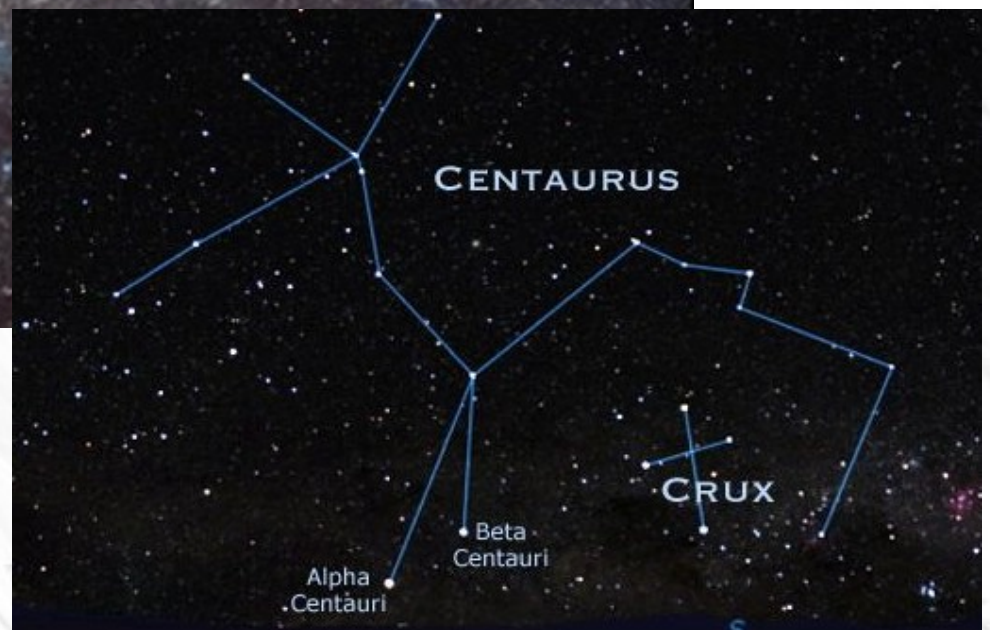
Example:

We see the Orion Nebula as it looked 1500 years ago.





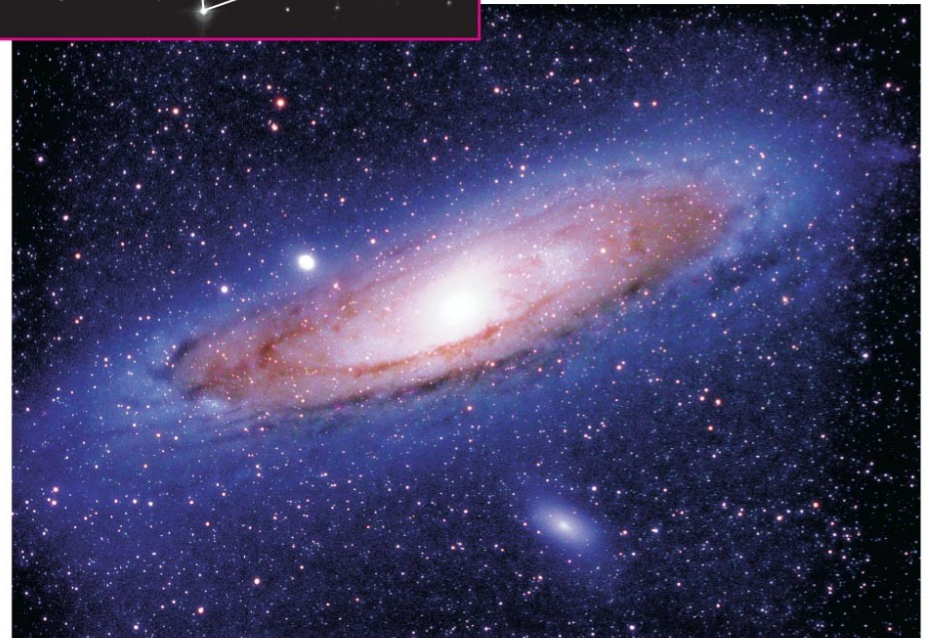
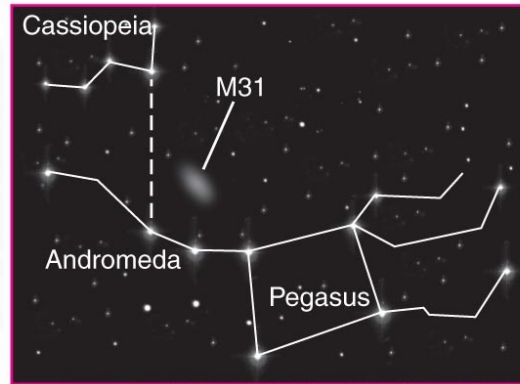
4.3 light years away



Example:

This photo shows the Andromeda Galaxy as it looked about 2 1/2 million years ago. Brightest galaxy we can see

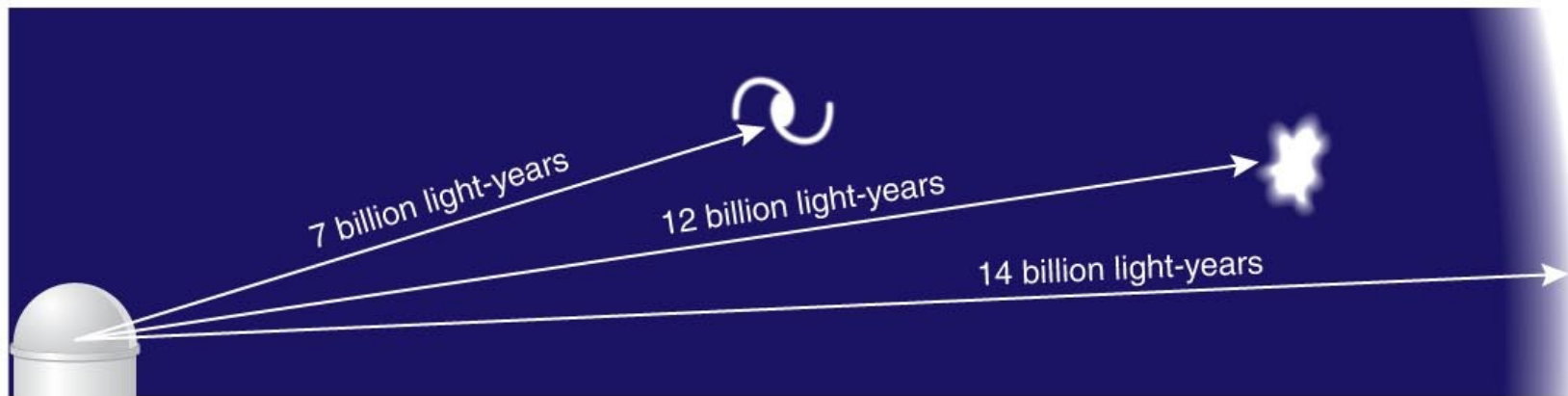
Question: When will be able to see what looks like now?



13.7 billion light years away



- At great distances, we see objects as they were when the universe was much younger.



Time Scales

- **Universe:** 13.7 billion years old (the expansion time since the “Big Bang,” the birth of the Universe)

$$1.37 \times 10^{10} \text{ yrs} = 4.3 \times 10^{17} \text{ seconds!}$$

- **Milky Way Galaxy:** about 13 billion years old (oldest stars are this age)
- **Solar System:** 4.6 billion years old (radioactive dating of meteorites)

More Time Scales

- **Unicellular life:** over 3.5 billion years old (oldest recognizable fossils)
- **“Cambrian explosion”:** about 550 million years ago (fossils of complex, hard-bodied animals)
example: trilobites
- **Dinosaur extinction:** 65 million years ago (extinction of 2/3 of life on Earth)
- **Early hominids:** about 4.5 million years ago (homo sapiens began about 160,000 years ago)

Stromatolites: oldest fossils. Can be found in Australia. Colonies of bacteria

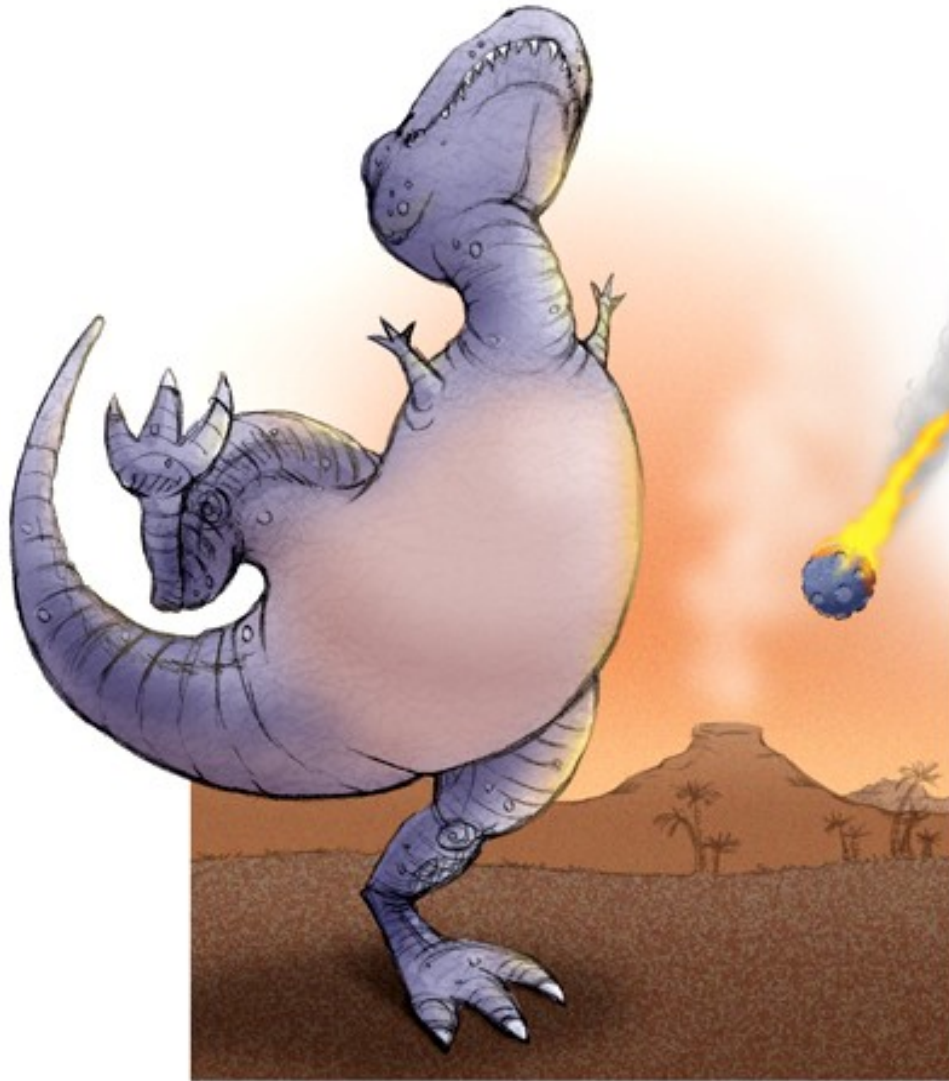


trilobite



Trilobites, exclusively marine animals, first appeared at the beginning of the Cambrian Period, about 542 million years ago, when they dominated the seas

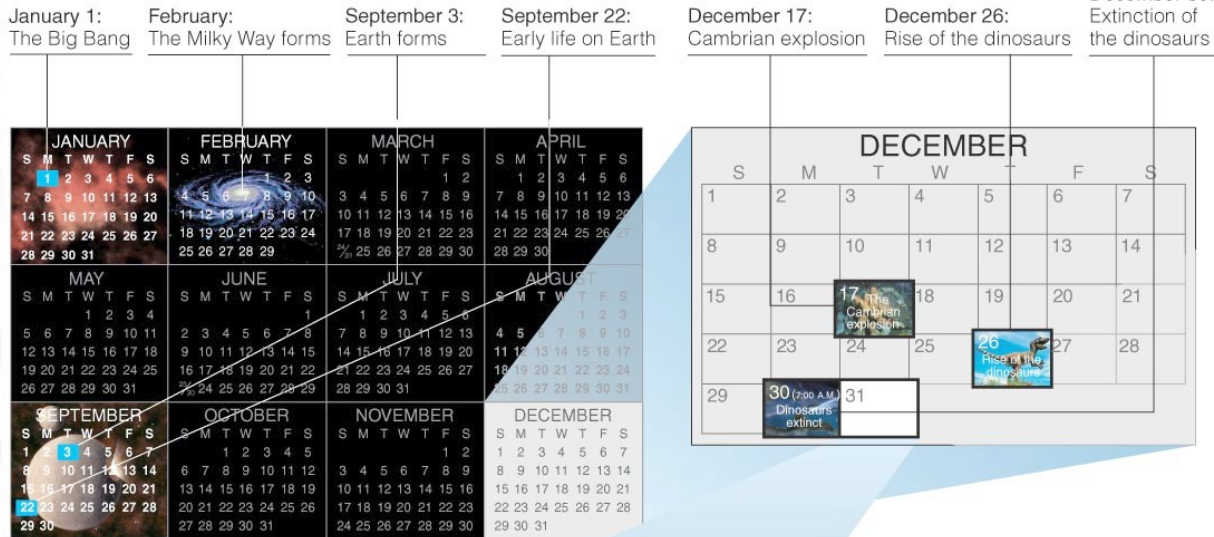
65 million years ago 2/3 of species was wiped out



Permian extinction = 250 million years ago

- The cosmic calendar: a scale on which we compress the history of the universe into 1 year.

THE HISTORY OF THE UNIVERSE IN 1 YEAR



December 31:

9:00 pm:
Early hominids evolve

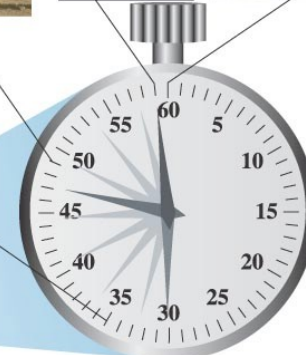
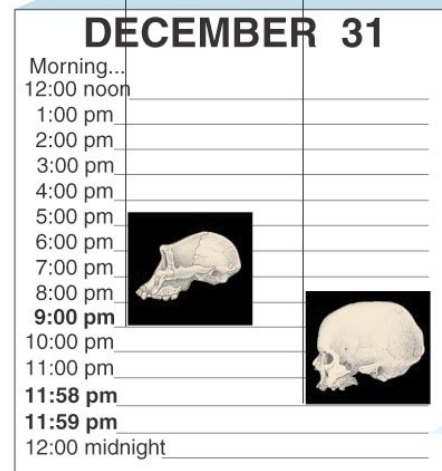
11:58 pm:
Modern humans evolve

25 seconds ago:
Agriculture arises

11 seconds ago:
Pyramids built

1 second ago:
Kepler and Galileo
show that Earth
orbits the Sun

Now



VIDEO
Big bang